What we have seen above is division on numbers. But how will you divide an algebraic expression by another algebraic expression?

Of course, the same procedure has to be followed for the algebraic expressions with the help of laws of exponents.

If x is a variable and m, n are constants, then $x^m \div x^n = x^{m-n}$ where $m > n$.

3.3.1 Division of a monomial by another monomial

Dividing a monomial $10p^4$ by another monomial $2p^3$, we get

$10p^4 \div 2p^3$

$$\frac{10p^4}{2p^3} = \frac{\overset{5}{\cancel{10}} \times p \times \cancel{p} \times \cancel{p} \times \cancel{p}}{\cancel{2} \times \cancel{p} \times \cancel{p} \times \cancel{p}} \quad \text{(expansion of power)}$$

$= 5p$

Think

Are the following correct?

(i) $\frac{x^3}{x^8} = x^{8-3} = x^5$

(ii) $\frac{10m^4}{10m^4} = 0$

(iii) When a monomial is divided by itself, we will get 1?

However, to divide we can also follow laws of exponents as,

$$\frac{\overset{5}{\cancel{10}}p^4}{\cancel{2}p^3} = 5p^{4-3}$$

$= 5p$

$$\frac{x^m}{x^n} = x^{m-n}$$

Example 3.6

Velu pastes '$4xy$' pictures in one page of his scrap book. How many pages will he need to paste $100x^2y^3$ pictures? (x, y are positive integers)



Solution:

Total number of pictures $= 100x^2y^3$

Pictures in one page $= 4xy$

Total number of pages needed $= \frac{\text{Total number of pictures}}{\text{pictures in one page}}$

$$= \frac{\overset{25}{\cancel{100}}x^2y^3}{\cancel{4}xy} = 25x^{2-1}y^{3-1}$$

$= 25xy^2$ pages

Try these

Divide

(i) $12x^3y^2$ by x^2y

(ii) $-20a^5b^2$ by $2a^3b^7$

(iii) $28a^4c^2$ by $21ca^2$

(iv) $(3x^2y)^3$ by $6x^2y^3$

(v) $64m^4(n^2)^3$ by $4m^2n^2$

(vi) $(8x^2y^2)^3$ by $(8x^2y^2)^2$

(vii) $81p^2q^4$ by $\sqrt{81p^2q^4}$

(vii) $(4x^2y^3)^0$ by $\frac{(x^3)^2}{x^6}$

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